

ROYAL FITNESS CLUB

Body Transformation Blueprint

MODULE 8

Final Assessment & Certificate

Duration	20 minutes
Course Code	CS_B3
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

ASSESSMENT INSTRUCTIONS

How to Complete This Assessment

- 25 questions covering all 7 modules of CS_B3 Body Transformation Blueprint
- One correct answer per question (A, B, C, or D)
- Passing score: 70% (18/25). Open-book assessment
- Target completion: 20 minutes

1. What body fat percentage range is classified as "fitness" for adult males?

- A) 3–5%
- B) 6–13%
- C) 14–17%
- D) 18–24%

2. Body recomposition is MOST achievable in which population?

- A) Advanced athletes with years of training
- B) Beginners in their first 1–2 years of resistance training
- C) Overweight individuals doing cardio only
- D) Athletes during a competition season peak

3. Lipolysis is the process of:

- A) Muscle protein synthesis stimulated by leucine
- B) Glycogen breakdown to glucose
- C) Triglycerides → fatty acids + glycerol (fat release from adipose tissue)
- D) Beta-oxidation of fatty acids in mitochondria

4. Chronic cortisol elevation preferentially deposits fat in which location?

- A) Gluteofemoral (hips and thighs)
- B) Subcutaneous fat (beneath skin everywhere)
- C) Intramuscular fat (IMTG)
- D) Visceral (abdominal, around organs)

5. Which statement about the "fat burning zone" is most accurate?

- A) Low-intensity exercise burns more total fat calories than high-intensity
- B) Low-intensity burns a higher fat percentage but high-intensity burns more total calories and absolute fat
- C) The fat burning zone does not exist
- D) Running at >75% HRmax burns 100% carbohydrates with no fat oxidation

6. Leptin is secreted by:

- A) The pancreas in response to high blood glucose
- B) The hypothalamus in response to caloric restriction
- C) Adipose tissue in proportion to fat stores
- D) The liver in response to physical exercise

7. Sleep restricting to 5.5 hours vs 8.5 hours during a caloric surplus results in (Nedeltcheva 2010):

- A) Identical body composition change between groups
- B) Only 18% of weight gained as muscle (vs 50% at 8.5h) — primarily fat gain in the sleep-deprived group
- C) Greater muscle gain in the sleep-deprived group due to anabolic adaptation
- D) No difference in hormones but slightly worse mood

8. The primary molecular reason the "interference effect" occurs is:

- A) Lactic acid from cardio impairs muscle protein synthesis directly
- B) AMPK (activated by endurance training) inhibits mTORC1 (required for hypertrophy)
- C) Cardio burns the glycogen needed for lifting weights
- D) Running causes micro-tears in leg muscles that compete for amino acids

9. To minimise concurrent training interference, cardio should ideally be separated from resistance training by:

- A) 30 minutes
- B) 1–2 hours
- C) At least 6 hours (ideally 24 hours or different days)
- D) No separation needed — do cardio first always

10. The Minimum Effective Volume (MEV) for hypertrophy per muscle group per week is approximately:

- A) 3–5 hard sets
- B) 6–9 hard sets
- C) 10–12 hard sets
- D) 20–25 hard sets

11. Which mechanism of hypertrophy is considered the primary driver (~60–70% of signal)?

- A) Metabolic stress / pump
- B) Muscle damage from eccentric loading
- C) Mechanical tension on actin-myosin crossbridges
- D) Growth hormone release during training

12. The hip thrust produces greater glute max activation than the squat by approximately:

- A) 5%

- B) 10%
- C) 30%+
- D) Equal activation — no difference

13. A 7-day average weight shows no change for 2 consecutive weeks during a fat loss programme. The FIRST adjustment should be:

- A) Immediately reduce calories by 500 kcal
- B) Add 2 HIIT sessions per week
- C) Increase NEAT by 2000 steps/day or reduce calories 100–200 kcal
- D) Switch to a ketogenic diet

14. Which of the following causes the most significant daily body weight fluctuation?

- A) Actual fat gain or loss
- B) Muscle protein synthesis changes
- C) High sodium meal causing water retention + glycogen storage
- D) Basal metabolic rate variation

15. Waist circumference above which measurement indicates HIGH metabolic risk in men?

- A) 80cm
- B) 88cm
- C) 94cm
- D) 102cm

16. NEAT (Non-Exercise Activity Thermogenesis) can vary between similar-sized individuals by up to:

- A) 100–200 kcal/day
- B) 300–500 kcal/day
- C) 500–1000 kcal/day
- D) 1500–2000 kcal/day

17. Sleep deprivation of 1 week (5h/night) reduces testosterone in males by approximately:

- A) 1–2%
- B) 5%
- C) 10–15%
- D) 25–30%

18. The optimal fat loss rate to preserve lean mass for most recreational athletes is:

- A) 0.1–0.2% body weight/week
- B) 0.5–1% body weight/week
- C) 1.5–2% body weight/week
- D) 2–3% body weight/week

19. Growth mindset vs fixed mindset: when a person hits a training plateau, growth mindset responds by:

- A) Concluding their genetics are the limiting factor
- B) Quitting the current programme immediately
- C) Treating the plateau as an adaptation signal and strategically adjusting variables
- D) Reducing training frequency to "avoid overtraining"

20. Which type of cardio is most appropriate during a hypertrophy block due to minimal muscle interference?

- A) 2× weekly HIIT running sessions
- B) 4× weekly sprint training
- C) 3–4× weekly LISS (low-intensity cycling, walking)
- D) Daily heavy rowing machine intervals

21. The Nedeltcheva 2010 study on sleep and body composition found that:

- A) Sleep duration does not affect body composition during caloric surplus
- B) Greater muscle gain occurred with 5.5h vs 8.5h sleep
- C) 8.5h sleep resulted in 50% of weight gained as lean mass vs 18% at 5.5h
- D) Sleep only affects recovery from endurance training, not resistance training

22. Which of the following is the most accurate statement about spot reduction?

- A) Ab exercises burn abdominal fat preferentially through increased local blood flow
- B) Spot reduction is scientifically proven for the abdominal region only
- C) Fat is mobilised systemically based on genetics/hormones — not adjacent to contracting muscle
- D) Spot reduction works for subcutaneous but not visceral fat

23. Ghrelin levels are most effectively managed by:

- A) Eating one large meal per day
- B) High protein intake + adequate sleep + high-volume vegetables
- C) Low-fat, high-carbohydrate meals before training
- D) Extended overnight fasting (>16 hours)

24. The 5-minute rule for low motivation days states:

- A) Rest is mandatory when motivation is below 5/10
- B) Commit to only 5 minutes of training, with permission to stop after
- C) 5 minutes of meditation before training replaces motivation
- D) 5 grams of caffeine should be consumed before training

25. NEAT is unconsciously suppressed by how many calories per day when dieting?

- A) 25–50 kcal/day
- B) 50–100 kcal/day
- C) 100–200 kcal/day

D) 200–400 kcal/day

ANSWER KEY

Q1: C	Q2: B	Q3: C	Q4: D	Q5: B
Q6: C	Q7: B	Q8: B	Q9: C	Q10: C
Q11: C	Q12: C	Q13: C	Q14: C	Q15: D
Q16: D	Q17: C	Q18: B	Q19: C	Q20: C
Q21: C	Q22: C	Q23: B	Q24: B	Q25: D

ROYAL FITNESS CLUB
BODY TRANSFORMATION BLUEPRINT CERTIFICATE

This certifies that

—
has completed the

Body Transformation Blueprint — CS_B3

demonstrating comprehensive knowledge of body composition, fat loss science, muscle building, cardiovascular training, progress tracking, transformation mindset, and lifestyle optimisation.

Instructor Signature

Date